SANDEEP AHIRWAR

Full-Stack Developer | Computer Vision & AI Engineer | DRDO Ex-Intern

sandeepverma.951978@gmail.com | (+91) 9519784281 | Lucknow UP, India | GitHub | LinkedIn

PROFESSIONAL SUMMARY

B.Tech ECE Engineer (CGPA: 8.5) with hands-on research experience at DRDO (IRDE). Expert in Computer Vision (YOLO, OpenCV) and Full-Stack Development (Node.js, React). Proven track record of building real-time detection systems and secure, scalable web applications. Targeting Full-Stack or AI/ML Engineering roles in high-growth environments.

TECHNICAL SKILLS

- **AI & Computer Vision:** YOLO (v5/v8), SSD, OpenCV, RAG, LangChain, FAISS, PyTorch, CNN, Feature Matching
- **Web Technologies:** Node.js, React.js, Express, REST APIs, JWT Authentication, RBAC
- **Languages & Databases:** Python, JavaScript, C, MySQL, MongoDB
- **Tools & Hardware:** Git, GitHub, Docker (Basic), ESP32, IoT Systems

PROFESSIONAL EXPERIENCE

DRDO (IRDE) – Dehradun, India | *Research Intern – Computer Vision & Detection* | Feb 2026 – Present

- **Object Detection Engineering:** Deployed YOLOv8 and SSD architectures for high-speed surveillance with real-time inference on live video feeds.
- **Pipeline Optimization:** Engineered custom OpenCV preprocessing pipelines to reduce false positives in low-light/noisy environments.
- **Model Evaluation:** Conducted comparative analysis of detection models balancing computational load and accuracy for defense applications.

Zidio Development | *Web Development Intern* | May 2024 – July 2024

- **System Architecture:** Built scalable backend modules using Node.js and React with secure JWT authentication and Role-Based Access Control (RBAC).

TECHNICAL PROJECTS

Real-Time Multi-Object Detection System | *Python, YOLO, OpenCV*

- Standalone system identifying and tracking multiple object classes from webcam inputs at 30+ FPS on edge devices.

Full-Stack E-Commerce Platform | *Node.js, Express, MySQL, React*

- Complete commercial engine with product management, secure JWT-based checkout, RBAC, and complex DB relationships.

LLM Programming Agent | *Python, StarCoder, SlimPajama*

- Fine-tuned LLMs for automated code generation with optimized response time for real-time programming logic suggestions.

Satellite Intelligence Pipeline | *Python, OpenCV, Image Processing*

- Detected urban structures in satellite imagery using edge detection and morphological filtering on multi-resolution datasets.

EDUCATION & CERTIFICATIONS

- **B.Tech – Electronics & Communication Engineering (CGPA: 8.5)** | 2022 – 2026
- **IIT Madras:** Certification in Embedded Systems & IoT
- **IIT Kanpur:** Certification in Metamaterial Antennas

AWARDS

- 3rd Place – National Level Project Exhibition (Engineer's Day 2024)